

1. NAME OF THE MEDICINAL PRODUCT

Firialta 10 mg film-coated tablets

Firialta 20 mg film-coated tablets

2. QUALITATIVE AND QUANTITATIVE COMPOSITION

Firialta 10 mg film-coated tablet

Each film-coated tablet contains 10 mg finerenone.

Firialta 20 mg film-coated tablet

Each film-coated tablet contains 20 mg finerenone.

3. PHARMACEUTICAL FORM

Film-coated tablet

Firialta 10 mg film-coated tablet

Pink, oval-oblong tablet with a length of 10 mm and a width of 5 mm, marked '10' on one side and 'FI' on the other side

Firialta 20 mg film-coated tablet

Pale-yellow, oval-oblong tablet with a length of 10 mm and a width of 5 mm, marked '20' on one side and 'FI' on the other side

4. CLINICAL PARTICULARS

4.1 Indication(s)

CKD T2D

Firialta is indicated to reduce the risk of sustained eGFR decline, end-stage kidney disease, cardiovascular death, nonfatal myocardial infarction, and hospitalization for heart failure in adult patients with chronic kidney disease (CKD) associated with type 2 diabetes (T2D).

Heart Failure (LVEF $\geq 40\%$)

Firialta is indicated to reduce the risk of cardiovascular death, hospitalization for heart failure, and urgent heart failure visits in adults with heart failure and left ventricular ejection fraction (LVEF) $\geq 40\%$.

4.2 Dosage and method of administration

4.2.1 Method of administration

Oral use

Tablets may be taken with a glass of water and with or without food (*see section 'Pharmacokinetic properties'*).

Avoid taking Firalta with grapefruit or grapefruit juice (*see section 'Special warnings and precautions for use' and '4.5 Interaction with other medicinal products and other forms of interaction'*).

For patients who are unable to swallow whole tablets, Firalta tablet may be crushed and mixed with water or soft foods, such as applesauce, immediately prior to use and administered orally (*see section 'Pharmacokinetic properties'*).

4.2.2 Dosage regimen

4.2.2.1 Initiation of treatment

Serum Potassium Limits

CKD T2D

In patients with CKD T2D, initiation of Firalta treatment is recommended when serum potassium ≤ 4.8 mmol/L. For monitoring of serum potassium, see 'Continuation of treatment.'

If serum potassium > 4.8 to 5.0 mmol/L, initiation of Firalta treatment may be considered with additional serum potassium monitoring within the first 4 weeks based on patient characteristics and serum potassium levels (*see section 'Special warnings and precautions for use'*).

If serum potassium > 5.0 mmol/L, initiation of Firalta treatment is not recommended (*see section 'Special warnings and precautions for use'*).

Heart Failure (LVEF $\geq 40\%$)

In patients with heart failure (LVEF $\geq 40\%$), initiation of Firalta treatment is recommended when serum potassium ≤ 5.0 mmol/L (*see section 'Special warnings and precautions for use'*).

Starting Dose for All Indications

Measure estimated glomerular filtration rate (eGFR) to determine the starting dose. The starting dose of Firalta is:

- 20 mg once daily if eGFR ≥ 60 mL/min/1.73 m²
- 10 mg once daily if eGFR ≥ 25 to < 60 mL/min/1.73 m²

Initiation of Firalta treatment is not recommended in patients with eGFR < 25 mL/min/1.73 m² as clinical experience is limited.

4.2.2.2 Target dose

CKD T2D

For patients with CKD T2D, the recommended target dose of Firalta is 20 mg once daily.

Heart Failure (LVEF $\geq 40\%$)

For patients with heart failure (LVEF $\geq 40\%$), the recommended target dose of Firalta depends on renal function (eGFR) at initiation of Firalta treatment (see Table 2):

- 40 mg once daily if eGFR ≥ 60 mL/min/1.73m²
- 20 mg once daily if eGFR ≥ 25 to < 60 mL/min/1.73m²

4.2.2.3 Continuation of treatment

Four weeks after initiation or re-start or dose adjustment of Firalta treatment, remeasure serum potassium and eGFR. See Table 1 or 2 to determine continuation of Firalta treatment and dose adjustment.

Thereafter, remeasure serum potassium periodically and as needed based on patient characteristics and serum potassium levels (*see section* ‘Special warnings and precautions for use’ *and* ‘4.5 Interaction with other medicinal products and other forms of interaction’).

Table 1: Continuation of Firalta Treatment and Dose Adjustment for Patients with CKD T2D

Serum potassium (mmol/L)	Firalta dose (after 4 weeks and thereafter)
≤ 4.8	Maintain 20 mg once daily. If on 10 mg once daily, increase to 20 mg once daily if eGFR has not decreased > 30% compared to prior measurement.
> 4.8 – 5.5	Maintain dose.
> 5.5	Withhold Firalta. Restart at 10 mg once daily if serum potassium ≤ 5.0 mmol/L.

Table 2: Continuation of Firalta Treatment and Dose Adjustment for Patients with Heart Failure (LVEF ≥40%)

Serum Potassium (mmol/L)	Firalta Dose (after 4 weeks and thereafter)
<5.0	If on 10 mg once daily, increase to 20 mg once daily.* If on 20 mg once daily, either maintain dose, or when eGFR ≥60 mL/min/1.73m ² at treatment initiation, increase to 40 mg once daily.* Maintain target dose of 40 mg once daily.
5.0 to <5.5	Maintain dose.
5.5 to <6.0	Decrease to next lower dose. Withhold Firalta if on 10 mg once daily. Restart at 10 mg once daily if serum potassium <5.5 mmol/L.
≥6.0	Withhold Firalta. Restart at 10 mg once daily if serum potassium <5.5 mmol/L.**
*if eGFR has not decreased >30% compared to prior measurement.	
**If repeated measurements ≥5.5 mmol/L, restart at 10 mg once daily when <5.0 mmol/L.	

4.2.2.4 Missed doses

A missed dose should be taken as soon as possible after it is noticed, but only on the same day. If this is not possible, the dose should be skipped, and the next dose taken as prescribed. Two doses should not be taken to make up for a missed dose.

4.2.2.5 Maximum daily dose

CKD T2D

For patients with CKD T2D, the maximum daily dose of Firalta is 20 mg.

Heart Failure (LVEF $\geq$ 40%)

For patients with heart failure (LVEF $\geq$ 40%), the maximum daily dose of Firalta is 40 mg.

4.2.3 Additional information on special populations

4.2.3.1 Patients with renal impairment

Initiation of Firalta treatment

In patients with $\text{eGFR} \geq 25$ to < 60 mL/min/1.73 m², the starting dose of Firalta is 10 mg once daily. See section 'Initiation of treatment.'

In patients with $\text{eGFR} < 25$ mL/min/1.73m², initiation of Firalta treatment is not recommended as clinical experience is limited (*see section 'Special warnings and precautions for use' and section 'Pharmacokinetic properties'*).

Continuation of Firalta treatment

In patients with mild, moderate or severe renal impairment, continue Firalta treatment and adjust dose based on serum potassium. Measure eGFR 4 weeks after initiation to determine up-titration. See Table 1 or 2 and section 'Continuation of treatment.'

In patients with end-stage renal disease ($\text{eGFR} < 15$ mL/min/1.73 m²), continue Firalta treatment with caution regarding serum potassium levels as clinical experience is limited (*see section 'Special warnings and precautions for use'*).

4.2.3.2 Patients with hepatic impairment

In patients with severe hepatic impairment (Child Pugh C), avoid treatment with Firalta (*see section 'Special warnings and precautions for use' and section 'Pharmacokinetic properties'*).

In patients with mild or moderate hepatic impairment, no initial dose adjustment is required (Child Pugh A or B) (*see section 'Pharmacokinetic properties'*).

In patients with moderate hepatic impairment (Child Pugh B), consider additional serum potassium monitoring and adapt monitoring according to patient characteristics (*see section 'Special warnings and precautions for use' and section 'Pharmacokinetic properties'*).

4.2.3.3 Patients taking concomitant medications

In patients taking Firalta concomitantly with moderate or weak CYP3A4 inhibitors, potassium supplements, trimethoprim, or trimethoprim-sulfamethoxazole, consider additional serum potassium monitoring and adapt monitoring according to patient characteristics and make Firalta treatment decisions as directed in Table 1 or 2. Temporary discontinuation of Firalta when taking trimethoprim, or trimethoprim-sulfamethoxazole, may be necessary (*see sections 'Special warnings and precautions for use' and 'Interaction with other medicinal products and other forms of interaction'*).

4.2.3.4 Pediatric patients

The safety and efficacy of Firalta have not been established in patients under 18 years of age. Therefore, Firalta is not recommended for use in pediatric patients.

4.2.3.5 Geriatric patients

No dose adjustment is required in the elderly (*see section 'Pharmacokinetic properties'*).

4.2.3.6 Sex

No dose adjustment is required based on sex (*see section 'Pharmacokinetic properties'*).

4.2.3.7 Body weight

No dose adjustment is required based on body weight (*see section 'Pharmacokinetic properties'*).

4.2.3.8 Race/Ethnicity

No dose adjustment is required based on race or ethnicity (*see section 'Pharmacokinetic properties'*).

4.2.3.9 Smoking status

No dose adjustment is required based on smoking status (*see section 'Pharmacokinetic properties'*).

4.3 Contraindications

Firalta is contraindicated in patients:

- taking concomitant medications that are strong CYP3A4 inhibitors (e.g., itraconazole, ketoconazole, ritonavir, nelfinavir, cobicistat, clarithromycin, telithromycin and nefazodone) (*see section 'Interaction with other medicinal products and other forms of interaction'*).
- with Addison's disease.

4.4 Special warnings and precautions for use

4.4.1 Hyperkalemia

Hyperkalemia has been observed in patients treated with Firalta.

Some patients are at a higher risk to develop hyperkalemia. Risk factors include low eGFR, higher serum potassium and previous episodes of hyperkalemia. Consider more frequent monitoring in these patients.

Initiation of Firalta treatment is not recommended if serum potassium > 5.0 mmol/L.

CKD T2D

In patients with CKD T2D, if serum potassium > 4.8 to 5.0 mmol/L, initiation of Firalta treatment may be considered with additional serum potassium monitoring within the first 4 weeks based on patient characteristics and serum potassium levels (*see section 'Dosage and method of administration'*).

In treated patients with CKD T2D, withhold Firalta if serum potassium > 5.5 mmol/L. Follow local guideline

for the management of hyperkalemia. Restart Firalta at 10 mg once daily if serum potassium ≤ 5.0 mmol/L (*see section 'Dosage and method of administration'*).

Heart Failure (LVEF $\geq 40\%$)

In treated patients with heart failure (LVEF $\geq 40\%$), withhold Firalta if serum potassium ≥ 6.0 mmol/L. Follow local guidelines for the management of hyperkalemia. Restart Firalta at 10 mg once daily if serum potassium < 5.5 mmol/L or if repeatedly ≥ 5.5 mmol/L, wait to restart until < 5.0 mmol/L (*see section 'Dosage and method of administration'*).

In all patients, remeasure serum potassium and eGFR 4 weeks after initiation or re-start or dose adjustment of Firalta treatment. Thereafter, remeasure serum potassium periodically and as needed based on patient characteristics and serum potassium levels (*see section 'Dosage and method of administration'*).

Concomitant medications

The risk of hyperkalemia also may increase with the intake of concomitant medications that may increase serum potassium (*see section 'Interaction with other medicinal products and other forms of interaction'*). See also 'Concomitant use of substances that affect finerenone exposure.'

Avoid concomitant use of Firalta with the following medications:

- potassium-sparing diuretics (e.g., amiloride, triamterene)
- other mineralocorticoid receptor antagonists (MRAs) (e.g., eplerenone, esaxerenone, spironolactone, canrenone)

Use Firalta with caution and monitor serum potassium when taken concomitantly with the following medications:

- potassium supplements
- trimethoprim, or trimethoprim-sulfamethoxazole. Temporary discontinuation of Firalta may be necessary.

4.4.2 Renal impairment

The risk of hyperkalemia increases with decreasing renal function. Ongoing monitoring of renal function should be performed as needed according to standard practice (*see section 'Dosage and method of administration' and section 'Pharmacokinetic properties'*).

Initiation of Firalta treatment is not recommended in patients with eGFR < 25 mL/min/1.73 m² as clinical experience is limited (*see section 'Dosage and method of administration'*).

Continue Firalta treatment with caution regarding serum potassium levels in patients with end-stage renal disease (eGFR < 15 mL/min/1.73 m²) as clinical experience is limited (*see section 'Dosage and method of administration'*).

4.4.3 Hepatic impairment

Patients with severe hepatic impairment (Child Pugh C) have not been studied (*see section 'Pharmacokinetic properties'*). Due to an expected significant increase in finerenone exposure, avoid use of Firalta in patients with severe hepatic impairment (*see section 'Dosage and method of administration'*).

Due to an increase in finerenone exposure, consider additional serum potassium monitoring and adapt monitoring according to patient characteristics in patients with moderate hepatic

impairment (Child Pugh B) (*see section 'Dosage and method of administration' and section 'Pharmacokinetic properties'*).

4.4.4 Concomitant use of substances that affect finerenone exposure

Moderate and weak CYP3A4 inhibitors

The concomitant use of Firalta with moderate CYP3A4 inhibitors (e.g., erythromycin and verapamil) and weak CYP3A4 inhibitors (e.g., amiodarone and fluvoxamine) is expected to increase finerenone exposure (*see section 'Interaction with other medicinal products and other forms of interaction'*). Monitor serum potassium especially during initiation of or changes to dosing of Firalta or the CYP3A4 inhibitor (*see section 'Dosage and method of administration'*).

Strong and moderate CYP3A4 inducers

Avoid concomitant use of Firalta with strong CYP3A4 inducers (e.g., rifampicin, carbamazepine, phenytoin, phenobarbital, St John's Wort) or moderate CYP3A4 inducers (e.g., efavirenz), which are expected to markedly decrease finerenone plasma concentrations and result in reduced therapeutic effect (*see section 'Interaction with other medicinal products and other forms of interaction'*).

Consider selection of an alternate concomitant medicinal product with no or weak potential to induce CYP3A4.

Grapefruit

Avoid concomitant intake of grapefruit or grapefruit juice as it is expected to increase the plasma concentration of finerenone (*see sections 'Dosage and method of administration' and 'Interaction with other medicinal products and other forms of interaction'*).

4.4.5 Embryo-fetal toxicity

Animal data have shown reproductive toxicity (*see section 'Preclinical safety data'*). The relevance for humans is unknown. Firalta should not be used during pregnancy unless there has been careful consideration of the benefit for the mother and the risk to the fetus. If the patient becomes pregnant while taking Firalta, the patient should be informed of potential risks to the fetus. Advise women of childbearing potential to use effective contraception during treatment with Firalta. Advise women not to breastfeed during treatment with Firalta (*see section 'Fertility, pregnancy and lactation'*).

4.5 Interaction with other medicinal products and other forms of interaction

4.5.1 Effects of other substances on finerenone

Finerenone is cleared almost exclusively via cytochrome P450 (CYP)-mediated oxidative metabolism (mainly CYP3A4 [90%] with a small contribution of CYP2C8 [10%]).

4.5.1.1 Effect of CYP3A4 inhibitors on finerenone

Strong CYP3A4 inhibitors

Simulations suggest that concomitant use of Firalta with itraconazole (200 mg BID), a strong CYP3A4 inhibitor, increases finerenone AUC (+531%) and C_{max} (+137%). Clarithromycin (500 mg BID), another strong inhibitor, also is predicted to increase finerenone AUC (+428%)

and C_{max} (+125%). Due to an expected marked increase in finerenone exposure, concomitant use of Firialta with itraconazole, clarithromycin and other strong CYP3A4 inhibitors (e.g., ketoconazole, ritonavir, nelfinavir, cobicistat, telithromycin or nefazodone) is contraindicated (*see section 'Contraindications'*).

Moderate CYP3A4 inhibitors

Concomitant use of erythromycin (500 mg thrice daily), a moderate CYP3A4 inhibitor, increased finerenone mean AUC and C_{max} by 248% and 88%, respectively. Another moderate CYP3A4 inhibitor, verapamil (240 mg controlled-release tablet once daily), increased finerenone mean AUC and C_{max} by 170% and 122%, respectively. Serum potassium may increase, and therefore, monitoring of serum potassium is recommended (*see sections 'Dosage and method of administration' and 'Special warnings and precautions for use'*).

Weak CYP3A4 inhibitors

In an analysis of Firialta in patients, the use of amiodarone, a weak CYP3A4 inhibitor, was estimated to result in a 21% increase of finerenone AUC. Simulations suggest that fluvoxamine (100 mg BID), another weak inhibitor, increases finerenone AUC (+57%) and C_{max} (+38%). Serum potassium may increase, and therefore, monitoring of serum potassium is recommended (*see sections 'Dosage and method of administration' and 'Special warnings and precautions for use'*).

Grapefruit

Concomitant intake of grapefruit or grapefruit juice is expected to increase the plasma concentration of finerenone and should be avoided (*see sections 'Dosage and method of administration' and 'Special warnings and precautions for use'*).

4.5.1.2 Effect of strong and moderate CYP3A4 inducers on finerenone

Simulations suggest that rifampicin (600 mg OD), a strong CYP3A4 inducer, decreases finerenone AUC (-93%) and C_{max} (-86%). Efavirenz (600 mg OD), a moderate CYP3A4 inducer, is predicted to decrease finerenone AUC (-81%) and C_{max} (-68%).

Concomitant use of Firialta with rifampicin and other strong CYP3A4 inducers (e.g., carbamazepine, phenytoin, phenobarbital, St John's Wort) or with efavirenz and other moderate CYP3A4 inducers, markedly decreases finerenone plasma concentration and results in reduced therapeutic effect and should be avoided (*see section 'Special warnings and precautions for use'*).

4.5.1.3 Lack of clinically relevant drug-drug interaction

Concomitant use of gemfibrozil (600 mg twice-daily), a strong inhibitor of CYP2C8, increased finerenone mean AUC and C_{max} by 10% and 16%, respectively. This is not clinically relevant.

Pre- and co-treatment with the proton pump inhibitor omeprazole (40 mg once daily) had no effect on finerenone mean AUC and mean C_{max}.

Concomitant use of antacid aluminum hydroxide and magnesium hydroxide (70 mVal) had no effect on finerenone mean AUC and reduced its mean C_{max} by 19%. This is not clinically relevant.

4.5.2 Effect of finerenone on other substances

At 40 mg once daily, finerenone is a weak inhibitor of the CYP3A4 enzyme *in vivo*. Co-administration of multiple doses of 40 mg finerenone with the CYP3A4 probe substrate midazolam resulted in a mean AUC increase of 31% with no effect on C_{max}. Consider the potential increased exposure of sensitive CYP3A4 substrates with a narrow therapeutic window in case of concomitant use of finerenone 40 mg once daily. A multiple-dose regimen of 20 mg finerenone once-daily had no effect on the AUC of the CYP3A4 probe substrate midazolam. Finerenone neither inhibits nor induces CYP3A4 at this dose level.

At 40 mg once daily, finerenone is a weak inhibitor of the CYP2C8 enzyme *in vivo*. Co-administration of multiple doses of 40 mg finerenone with the CYP2C8 probe substrate repaglinide resulted in a mean AUC and C_{max} increase of 59% and 30%, respectively. Consider the potential increased exposure of sensitive CYP2C8 substrates with a narrow therapeutic window in case of concomitant use of finerenone 40 mg once daily. A single dose of 20 mg finerenone had no effect on AUC and C_{max} of the CYP2C8 probe substrate repaglinide. Finerenone does not inhibit CYP2C8 at this dose level.

Lack of mutual pharmacokinetic interaction was demonstrated between finerenone and the CYP2C9 substrate warfarin. Finerenone had no effect on AUC and C_{trough} of the P-gp substrate digoxin at steady-state (90% CI of ratio (digoxin + finerenone / digoxin alone) within 90-111%). The ratio of digoxin C_{max} at steady-state (digoxin + finerenone / digoxin alone) was 105.3% with a 90% CI of 94.65-117.16 %.

Multiple doses of 40 mg finerenone once daily had no clinically relevant effect on AUC or C_{max} of the BCRP and OATP substrate rosuvastatin.

4.5.3 Pharmacodynamic interactions

Medications that increase serum potassium

It is anticipated that medications that increase serum potassium will increase the risk of hyperkalemia when used concomitantly with Firalta.

Concomitant use of Firalta with the following medications should be avoided:

- potassium-sparing diuretics (e.g., amiloride, triamterene)
- other mineralocorticoid receptor antagonists (MRAs) (e.g., eplerenone, esaxerenone, spironolactone, canrenone)

Firalta should be used with caution and serum potassium monitored when taken concomitantly with the following medications:

- potassium supplements
- trimethoprim, or trimethoprim - sulfamethoxazole. Temporary discontinuation of Firalta may be necessary.

(See section 'Special warnings and precautions for use')

4.6 Fertility, pregnancy and lactation

4.6.1 Pregnancy

There are no data on the use of Firalta in pregnant women . Animal studies have shown embryo- fetal developmental toxicity at exposures in excess to the maximum human exposure. In the pre- and post-natal developmental toxicity study, slightly increased locomotor activity was found in the offspring, which may have been caused by exposure during pregnancy (*see section 'Preclinical safety data'*).

Firialta should not be used during pregnancy unless there has been careful consideration of the benefit for the mother and the risk to the fetus (*see section 'Special warnings and precautions for use'*).

4.6.2 Lactation

It is unknown whether finerenone or its metabolites are excreted in human breast milk. Available pharmacokinetic and toxicological data in animals have shown excretion of finerenone and its metabolites in milk. Rat pups exposed by this route showed adverse effects. A risk to the nursing infant cannot be excluded. Breastfeeding should be discontinued if use of Firialta is considered essential (*see section 'Special warnings and precautions for use'*).

4.6.3 Fertility

No human data on the effect of Firialta on fertility is available. Animal studies with finerenone did not indicate a risk of impaired male fertility. Animal studies with finerenone indicated impaired female fertility at exposures considered sufficiently in excess to the maximum human exposure indicating no clinical relevance (*see section 'Preclinical safety data'*).

4.6.4 Women of childbearing potential / Contraception

Firialta may cause embryo-fetal harm when administered during pregnancy. Women of childbearing potential should use effective contraception during treatment with Firialta (*see section 'Special warnings and precautions for use'*).

4.7 Effects on ability to drive or use machines

No effects on ability to drive and use machines have been observed.

4.8 Undesirable effects

4.8.1 Summary of the safety profile

The safety of Firialta in patients with chronic kidney disease associated with type 2 diabetes was evaluated in two pivotal phase III studies, FIDELIO-DKD and FIGARO-DKD. In the FIDELIO-DKD study, 2,818 patients received Firialta (10 or 20 mg once daily) with a mean duration of treatment of 2.2 years. In the FIGARO-DKD study, 3,671 patients received Firialta (10 mg or 20 mg once daily) with a mean duration of treatment of 2.9 years.

The safety of Firialta in patients with heart failure (LVEF $\geq 40\%$) was evaluated in the phase III study, FINEARTS-HF. In this study, 2,993 patients received Firialta (10 mg, 20 mg, or 40 mg once daily) with a mean duration of treatment of 2.1 years.

The most frequently reported ($\geq 10\%$) adverse reaction was hyperkalemia. See 'Description of selected adverse reactions' below (*see section 'Special warnings and precautions for use'*).

4.8.2 Tabulated list of adverse reactions

The adverse reactions reported with Firialta are summarized in Table 3 below by MedDRA system organ class and by frequency.

Adverse reactions are ranked by System organ class and then by frequency with the most frequent first, using the following convention:

very common ($\geq 1/10$)

common ($\geq 1/100$ to $< 1/10$)

uncommon ($\geq 1/1,000$ to $< 1/100$)

rare ($\geq 1/10,000$ to $< 1/1,000$)

very rare ($< 1/10,000$)

Within each frequency grouping, adverse reactions are ranked in order of decreasing seriousness.

Table 3: Adverse Reactions Reported with Firalta in Phase III Studies (Pooled FIDELIO- DKD, FIGARO-DKD, and FINEARTS-HF)

MedDRA System Organ Class	Very common	Common
Metabolism and nutrition disorders	Hyperkalemia ¹	Hyponatremia ² Hyperuricemia ^{3,4}
Vascular disorders		Hypotension ^{5,6}
Investigations		Blood creatinine increased ⁷ / Glomerular filtration rate decreased ⁷

¹ includes Blood potassium increased and Hyperkalemia

² includes Blood sodium decreased and Hyponatremia

³ includes Blood uric acid increased and Hyperuricemia

⁴ Asymptomatic hyperuricemia was observed. In the FIGARO-DKD study, an increase from baseline in mean serum uric acid of up to 0.3 mg/dL was seen in the Firalta group compared to placebo, which attenuated over time. No hyperuricemia related treatment discontinuations were reported.

⁵ includes Blood pressure decreased, Blood pressure diastolic decreased, Diastolic hypotension, and Hypotension

⁶ In patients treated with Firalta, the mean systolic blood pressure (SBP) decreased by 3 mmHg and the mean diastolic blood pressure (DBP) decreased by 1-2 mmHg at month 1, remaining stable thereafter. The majority of hypotension events were mild or moderate and resolved. Events associated with hypotension, e.g., dizziness, syncope, or fall, were comparable between Firalta and placebo.

⁷ An initial small increase in blood creatinine and decrease in glomerular filtration rate (GFR) (mean 2-3 mL/min/1.73m²) occurred within the first 4 weeks of starting Firalta therapy, and then stabilized. These changes were reversible after treatment discontinuation.

4.8.3 Description of selected adverse reactions

Hyperkalemia

In the FIDELIO-DKD study, including patients with CKD (mean eGFR 44.4 mL/min/1.73 m²) and T2D, hyperkalemia events were reported in 18.2% of Firalta-treated patients compared with 9.0% of placebo-treated patients. An increase from baseline in mean serum potassium in the first month of treatment of approximately 0.2 mmol/L was observed in the Firalta group compared to placebo, which remained stable thereafter. In the FIGARO-DKD study, including patients with CKD (mean eGFR 67.8 mL/min/1.73 m²) and T2D, hyperkalemia events were

reported in 10.7% of Firalta- treated patients compared with 5.3% of placebo-treated patients. An increase from baseline in mean serum potassium in the first month of treatment of approximately 0.15 mmol/L was observed in the Firalta group compared to placebo, which remained stable thereafter.

In the FINEARTS-HF study, including patients with heart failure (LVEF $\geq 40\%$), hyperkalemia events were reported in 9.7% of Firalta-treated patients compared with 4.2% of placebo-treated patients. An increase from baseline in mean serum potassium in the first month of treatment of approximately 0.2 mmol/L was observed in the Firalta group compared to placebo, which remained stable thereafter.

In all studies, the majority of hyperkalemia events were mild to moderate in patients treated with Firalta. For specific recommendations, refer to sections ‘Dosage and method of administration’ and ‘Special warnings and precautions for use.’

4.9 Overdose

No cases of adverse events associated with finerenone overdose in humans have been reported. The most likely manifestation of overdose is anticipated to be hyperkalemia. If hyperkalemia develops, standard treatment should be initiated.

Finerenone is unlikely to be efficiently removed by hemodialysis given its fraction bound to plasma proteins of about 90%.

5. PHARMACOLOGICAL PROPERTIES

5.1 Pharmacodynamic properties

Pharmacotherapeutic group: aldosterone antagonists

ATC Code: C03DA05

5.1.1 Mechanism of action

Finerenone is a nonsteroidal, selective antagonist of the mineralocorticoid receptor (MR) that potently attenuates inflammation and fibrosis mediated by MR overactivation. The MR is expressed in the kidneys, heart and blood vessels where finerenone also counteracts sodium retention and hypertrophic processes. Finerenone has a high potency and selectivity for the MR due to its nonsteroidal structure and bulky binding mode. Finerenone has no relevant affinity for androgen, progesterone, estrogen and glucocorticoid receptors and therefore does not cause sex hormone-related adverse events (e.g., gynecomastia). Its binding to the MR leads to a specific receptor ligand complex that blocks recruitment of transcriptional coactivators implicated in the expression of pro-inflammatory and pro-fibrotic mediators.

5.1.2 Pharmacodynamic effects

5.1.2.1 Effects on UACR

In FIDELIO-DKD and FIGARO-DKD, randomized, double-blind, placebo-controlled, multicenter phase III studies in adults with CKD and T2D, the placebo-corrected relative reduction in urinary albumin-to-creatinine ratio (UACR) in patients randomized to finerenone at month 4 was 31% and 32%, respectively, and UACR remained reduced throughout both studies.

In FINEARTS-HF, a randomized, double-blind, placebo-controlled, multicenter phase III study in adults with heart failure (LVEF $\geq 40\%$), the placebo-corrected relative reduction in UACR in patients randomized to finerenone at month 6 was 30%, and UACR remained reduced up to the last measurement at year 2.

In ARTS-DN, a randomized, double-blind, placebo-controlled, multicenter phase IIb dose-finding study in adults with CKD and T2D, the placebo-corrected relative reduction in UACR at day 90 was 25% and 38% in patients treated with finerenone 10 mg and 20 mg once daily, respectively.

5.1.2.2 Cardiac electrophysiology

In a thorough QT study in 57 healthy participants, there was no indication of a QT/QTc prolonging effect of finerenone after single doses of 20 mg (therapeutic) or 80 mg (supratherapeutic), indicating that finerenone has no effect on cardiac repolarization.

5.1.3 Clinical efficacy and safety

5.1.3.1 Chronic kidney disease associated with type 2 diabetes

Firialta was investigated in two randomized, double-blind, placebo-controlled, multicenter phase III studies, FIDELIO-DKD and FIGARO-DKD. In these studies, the effect of Firialta on kidney and cardiovascular outcomes was evaluated in adults with CKD and T2D receiving either Firialta 10 mg or 20 mg once daily, or placebo.

In FIDELIO-DKD patients were eligible based on evidence of persistent albuminuria (≥ 30 mg/g to < 300 mg/g) and $\text{eGFR} \geq 25$ but < 60 mL/min/1.73 m² and presence of diabetic retinopathy or persistent albuminuria (≥ 300 mg/g to 5,000 mg/g) and $\text{eGFR} \geq 25$ to < 75 mL/min/1.73 m², serum potassium ≤ 4.8 mmol/L at screening, and were required to be receiving standard of care, including a maximum tolerated labeled dose of an angiotensin-converting enzyme inhibitor (ACEi) or angiotensin receptor blocker (ARB). The trial excluded patients with known significant non-diabetic kidney disease.

Patients with a clinical diagnosis of chronic heart failure with reduced ejection fraction and persistent symptoms (New York Heart Association class II to IV) were excluded.

The primary endpoint in the FIDELIO-DKD study was a composite of time to first occurrence of kidney failure (defined as chronic dialysis or kidney transplantation, or a sustained decrease in eGFR to < 15 mL/min/1.73 m² over at least 4 weeks), a sustained decline in eGFR of 40% or more compared to baseline over at least 4 weeks, or renal death. The key secondary endpoint was a composite of time to first occurrence of cardiovascular (CV) death, non-fatal myocardial infarction (MI), non-fatal stroke or hospitalization for heart failure.

The study analyzed 5,662 patients randomly assigned to receive either Firialta (N=2824), or placebo (N=2838), with a median follow-up duration of 2.6 years. After the end of study notification, vital status was obtained for 99.7% of patients. The trial population was 63% White, 25% Asian and 5% Black. The mean age at enrollment was 66 years and 70% of patients were male. At baseline, the mean eGFR was 44.4 mL/min/1.73 m², with 55% of patients having an $\text{eGFR} < 45$ mL/min/1.73 m², median urine albumin-to-creatinine ratio (UACR) was 853 mg/g, and mean glycated hemoglobin A1c (HbA1c) was 7.7%, 46% had a history of atherosclerotic cardiovascular disease, 30% had history of coronary artery disease, 8% had a history of cardiac failure, and the mean blood pressure was 138/76 mmHg. The mean duration of type 2 diabetes at baseline was 16.6 years and a history of diabetic retinopathy and diabetic neuropathy was reported in 47% and 26% of patients, respectively. At baseline, almost all patients were on ACEi (34%) or ARB (66%), and 97% of patients used one

or more antidiabetic medications (insulin [64%], biguanides [44%], glucagon-like peptide-1 [GLP-1] receptor agonists [7%], sodium-glucose cotransporter 2 [SGLT2] inhibitors [5%]). The other most frequent medications taken at baseline were statins (74%) and calcium channel blockers (63%).

Firialta demonstrated superiority to placebo by significantly reducing the risk of the primary composite endpoint compared to placebo in a time-to-event analysis using the Cox proportional hazards model and log-rank test (HR 0.82, 95% CI 0.73-0.92, $p=0.0009$). See Figure 1/Table 4 below. Firialta also significantly reduced the risk of the key secondary composite endpoint of time to first occurrence of CV death, non-fatal MI, non-fatal stroke or hospitalization for heart failure compared to placebo (HR 0.86, 95% CI 0.75-0.99, $p=0.0344$). See Figure 2. Prespecified secondary time-to-event endpoints are included in Table 4. The treatment effect for the primary and key secondary endpoints was generally consistent across subgroups, including region, eGFR, UACR, systolic blood pressure (SBP) and HbA1c at baseline.

In the FIDELIO-DKD study, hyperkalemia events were reported in 18.2% of Firialta-treated patients compared with 9.0% of placebo-treated patients. Hospitalization due to hyperkalemia for the Firialta group was 1.4% versus 0.3% in the placebo group. Hyperkalemia leading to permanent discontinuation in patients who received Firialta was 2.3% versus 0.9% in the placebo group.

In the FIDELIO-DKD study, Glomerular filtration rate decreased events were reported in 6.4% of Firialta-treated patients compared with 4.7% of placebo-treated patients, and those leading to permanent discontinuation in patients receiving Firialta were 0.2% versus 0.3% in the placebo group. Patients on Firialta experienced an initial decrease in eGFR (mean 2 mL/min/1.73m²) that attenuated over time compared to placebo. This decrease was reversible after treatment discontinuation. The initial decrease in eGFR was associated with long term preservation of kidney function.

The FIGARO-DKD study included adults with CKD and T2D, based on having a UACR of ≥ 30 mg/g to <300 mg/g and an eGFR of 25 to 90 mL/min/1.73m², or a UACR ≥ 300 mg/g and an eGFR ≥ 60 mL/min/1.73m² at screening. Patients were required to have a serum potassium of ≤ 4.8 mmol/L at screening and received standard of care, including a maximum tolerated labeled dose of a RAS inhibitor (either an ACEi or ARB).

The primary endpoint in the FIGARO-DKD study was a composite of time to first occurrence of CV death, non-fatal MI, non-fatal stroke or hospitalization for heart failure. Secondary endpoints included a composite of time to kidney failure, a sustained decline in eGFR of 40% or more compared to baseline over at least 4 weeks, or renal death and a composite of time to kidney failure, a sustained decline in eGFR of 57% or more compared to baseline, or renal death.

The study analyzed 7,328 patients randomly assigned to receive either Firialta (N=3674), or

placebo (N=3654) that were followed for a median duration of 3.4 years. After the end of study notification, vital status was obtained for 99.8% of patients. The trial population was 72% White, 20% Asian and 4% Black. The mean age at enrollment was 64 years and 69% of patients were male. At baseline, the mean eGFR was 67.8 mL/min/1.73 m², with 62% of patients having an eGFR $\geq$ 60 mL/min/1.73 m², median UACR was 309 mg/g, and mean glycated HbA1c was 7.7%, 45% of patients had a history of atherosclerotic cardiovascular disease, 8% had a history of cardiac failure, and the mean blood pressure was 136/77 mmHg. The mean duration of type 2 diabetes at baseline was 14.5 years and a history of diabetic retinopathy and diabetic neuropathy was reported in 31% and 28% of patients, respectively. At baseline, almost all patients were on a RAS-inhibitor and 98% of patients used one or more antidiabetic medications (insulin [54%], biguanides [69%], GLP-1 receptor agonists [8%], SGLT2 inhibitors [8%]). The other most frequent medication class taken at baseline was statins (71%).

Firialta significantly reduced the risk of the primary composite endpoint compared to placebo in a time to event analysis using the Cox proportional hazards model and log rank test (HR 0.87, 95% CI 0.76-0.98, p=0.0254). See Figure 3 and Table 5 below. The treatment effect for the primary endpoint was consistent across subgroups, including region, eGFR, UACR, SBP and HbA1c at baseline. A lower incidence rate of the secondary composite outcome of kidney failure, sustained eGFR decline of 40% or more or renal death was observed in the Firialta group compared to placebo, however this difference did not achieve statistical significance (HR 0.87, 95% CI 0.75-1.01, p=0.0635). See Figure 4 and Table 5 below. A lower risk of the secondary outcome of kidney failure, sustained eGFR decline of 57% or more or renal death was observed in the Firialta group compared to placebo (HR 0.77, 95% CI 0.60-0.99). Prespecified secondary time-to-event endpoints are included in Table 5.

In the FIGARO-DKD study, hyperkalemia events were reported in 10.7% of Firialta-treated patients compared with 5.3% of placebo-treated patients. Hospitalization due to hyperkalemia for the Firialta group was 0.6% versus <0.1% in the placebo group. Hyperkalemia leading to permanent discontinuation in patients who received Firialta was 1.3% versus 0.4% in the placebo group.

In the FIGARO-DKD study, Glomerular filtration rate decreased events were reported in 4.6% of Firialta-treated patients compared with 3.9% of placebo-treated patients, and those leading to permanent discontinuation in patients receiving Firialta were 0.2% versus 0.1% in the placebo group. Patients on Firialta experienced an initial decrease in eGFR of around 2 mL/min/1.73m² that attenuated over time compared to placebo. This decrease was reversible after treatment discontinuation. The initial decrease in eGFR was associated with long term preservation of kidney function.

Table 4: Analysis of the Primary and Secondary Time-to-Event Endpoints (and their Individual Components) in Phase III Study FIDELIO-DKD

	Subjects with Chronic Kidney Disease and Type 2 Diabetes					
	Firialta* 10 or 20 mg OD N=2824		Placebo* N=2838		Treatment Effect Firialta / Placebo	
Primary and Secondary Time-to-event Endpoints:	n (%)	Event Rate (100 pt-yr)	n (%)	Event Rate (100 pt-yr)	Hazard Ratio [95% CI]	p-value
Primary composite of kidney failure, sustained eGFR decline ≥40% or renal death	498 (17.6%)	7.53	600 (21.1%)	9.09	0.82 [0.73; 0.92]	0.0009
Kidney failure	205 (7.3%)	2.96	235 (8.3%)	3.39	0.86 [0.72; 1.05]	-
Sustained eGFR decline ≥40%	473 (16.7%)	7.15	577 (20.3%)	8.74	0.81 [0.72; 0.91]	-
Renal death	2 (<0.1%)	-	2 (<0.1%)	-	-	-
Secondary composite of CV death, non-fatal MI, non-fatal stroke or hospitalization for heart failure	366 (13.0%)	5.11	420 (14.8%)	5.93	0.86 [0.75; 0.99]	0.0344
CV death	128 (4.5%)	1.70	150 (5.3%)	1.99	0.86 [0.68;1.09]	-
Non-fatal MI	70 (2.5%)	0.94	87 (3.1%)	1.18	0.80 [0.58;1.09]	-
Non-Fatal stroke	90 (3.2%)	1.22	87 (3.1%)	1.18	1.03 [0.77;1.38]	-

Hospitalization for heart failure	138 (4.9%)	1.88	162 (5.7%)	2.22	0.85 [0.68;1.07]	-
All-cause mortality	219 (7.8%)	2.90	244 (8.6%)	3.24	0.90 [0.75;1.08]	0.2348**
All-cause hospitalization	1259 (44.6%)	22.59	1321 (46.5%)	23.91	0.95 [0.88; 1.02]	-
Kidney failure, sustained eGFR decline $\geq 57\%$ or renal death	248 (8.8%)	3.60	326 (11.5%)	4.74	0.75 [0.65; 0.90]	-

* Treatment in addition to maximum tolerated labeled doses of ACEi or ARB.

** Not significant

Figure 1: Time to first occurrence of kidney failure, sustained decline in eGFR $\geq 40\%$ from baseline, or renal death in the FIDELIO-DKD study

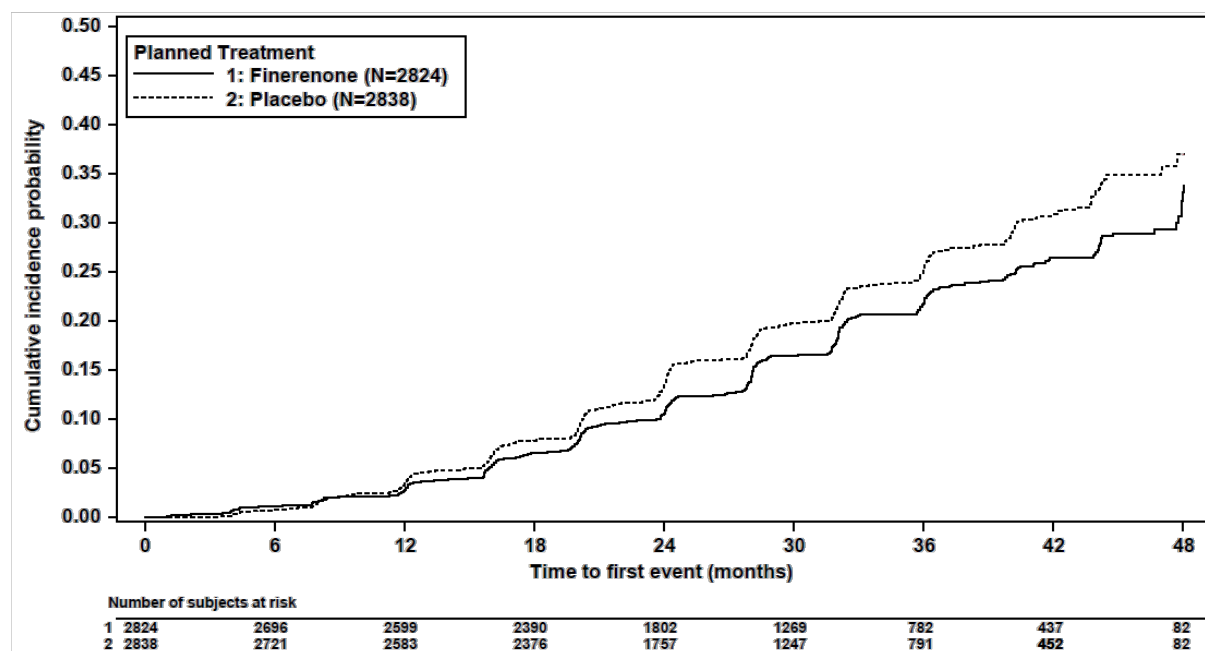


Figure 2: Time to first occurrence of CV death, non-fatal myocardial infarction, non-fatal stroke or hospitalization for heart failure in the FIDELIO-DKD study

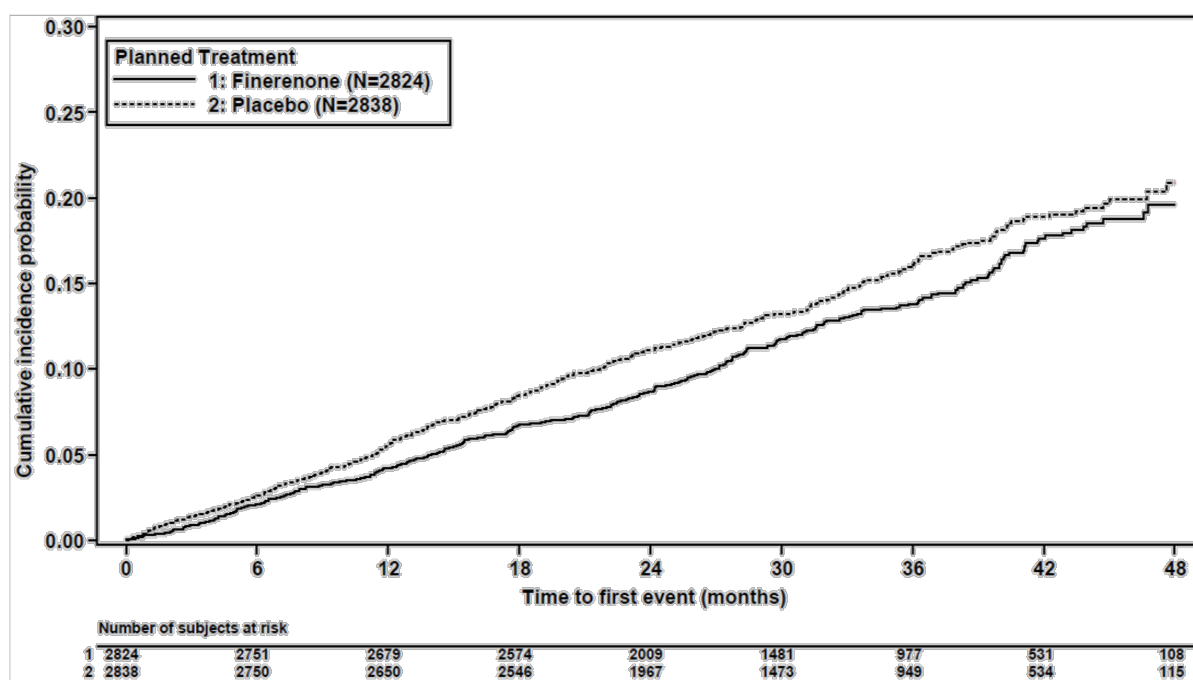


Table 5: Analysis of the Primary and Secondary Time-to-Event Endpoints (and their Individual Components) in Phase III Study FIGARO-DKD

	Subjects with Chronic Kidney Disease and Type 2 Diabetes					
	Firialta* 10 or 20 mg OD N=3674		Placebo* N=3654		Treatment Effect Firialta / Placebo	
Primary and Secondary Time-to-event Endpoints:	n (%)	Event Rate (100 pt-yr)	n (%)	Event Rate (100 pt-yr)	Hazard Ratio [95% CI]	p-value
Primary composite of CV death, non-fatal MI, non-fatal stroke or hospitalization for heart failure	457 (12.4%)	3.88	518 (14.2%)	4.46	0.87 [0.76; 0.98]	0.0254
CV death	193 (5.3%)	1.56	214 (5.9%)	1.75	0.89 [0.73; 1.08]	-
Non-fatal MI	103 (2.8%)	0.85	101 (2.8%)	0.84	1.00 [0.76; 1.32]	-
Non-fatal stroke	108 (2.9%)	0.89	111 (3.0%)	0.93	0.97 [0.74; 1.26]	-
Hospitalization for heart failure	117 (3.2%)	0.97	163 (4.5%)	1.36	0.71[0.56; 0.90]	-
Composite of kidney failure, sustained eGFR decline $\geq 40\%$ or renal death	350 (9.5%)	3.17	395 (10.8%)	3.59	0.87 [0.75; 1.01]	0.0635* *
Kidney failure	46 (1.3%)	0.40	62 (1.7%)	0.55	0.72 [0.49; 1.05]	-

Sustained eGFR decline $\geq 40\%$	338 (9.2%)	3.06	385 (10.5%)	3.50	0.86 [0.74; <1.00]	-
Renal death	0	-	2 (<0.1%)	-	-	-
All-cause hospitalization	1569 (42.7%)	16.94	1599 (43.8%)	17.54	0.97 [0.90; 1.04]	-
All-cause mortality	332 (9.0%)	2.69	370 (10.1%)	3.03	0.89 [0.77; 1.03]	-
Composite of kidney failure, sustained eGFR decline $\geq 57\%$ or renal death	108 (2.9%)	0.95	139 (3.8%)	1.23	0.77 [0.60; 0.99]	-

* Treatment in addition to maximum tolerated labeled doses of ACEi or ARB.

** Not significant

Figure 3: Time to first occurrence of CV death, non-fatal myocardial infarction, non-fatal stroke or hospitalization for heart failure in the FIGARO-DKD study

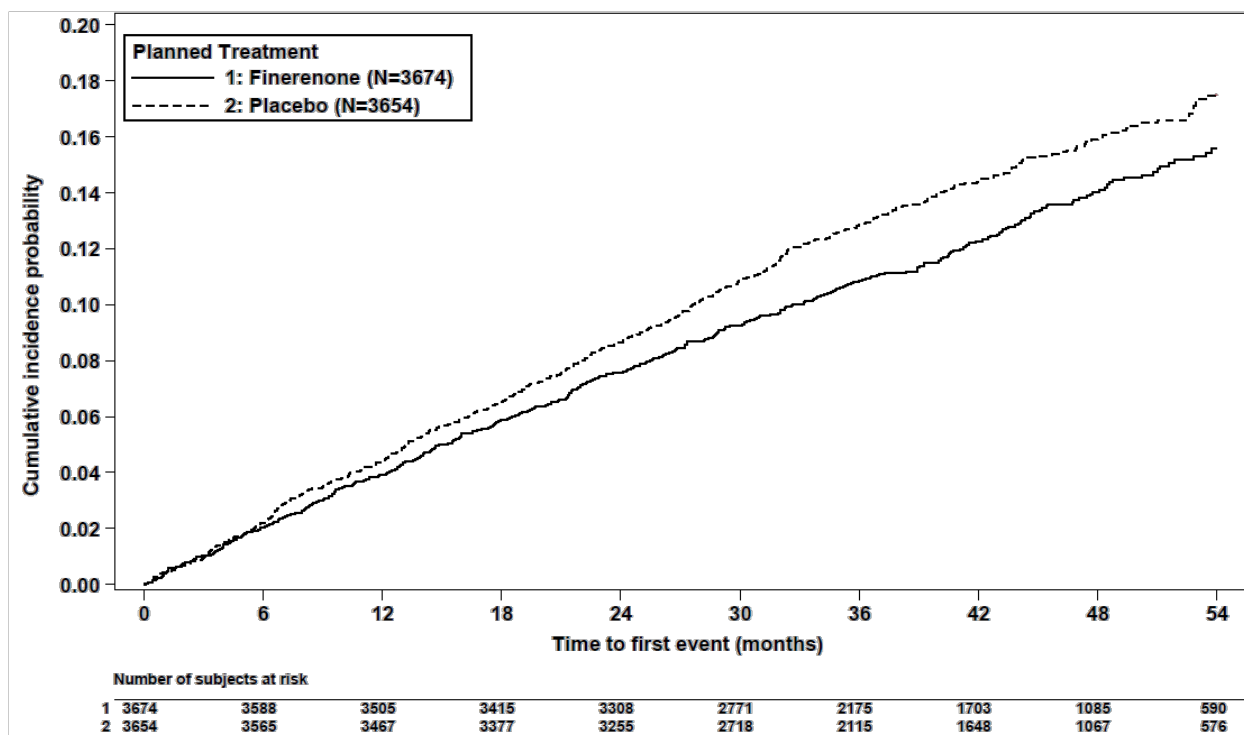
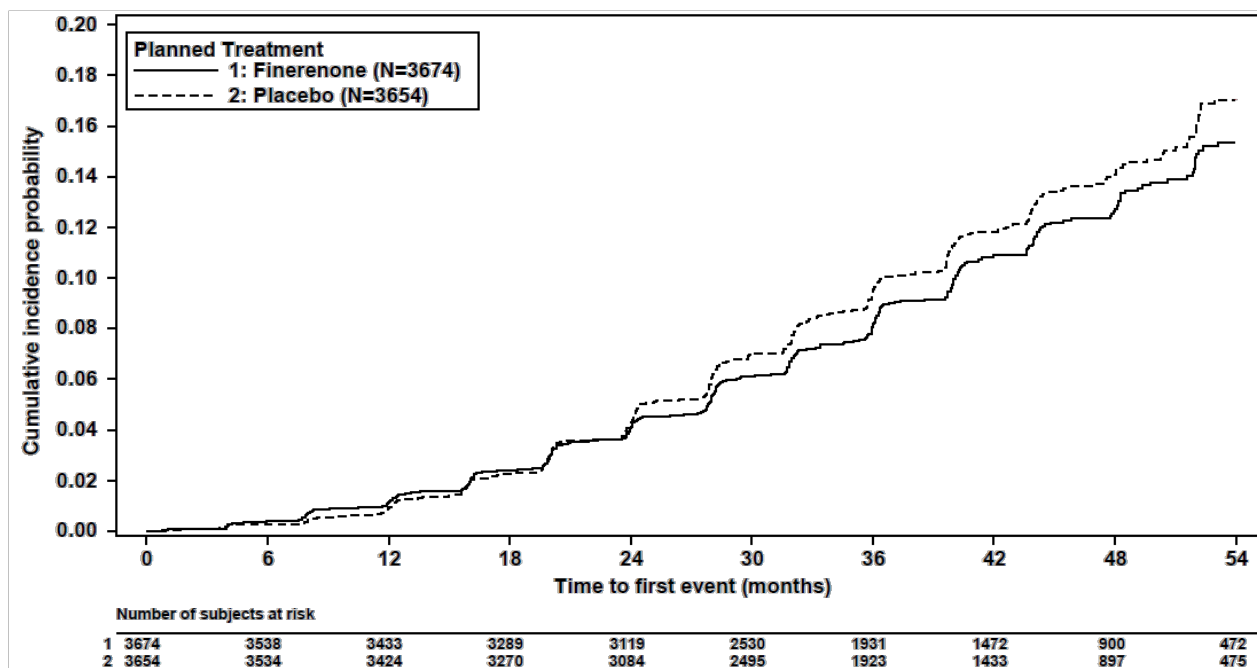


Figure 4: Time to first occurrence of kidney failure, sustained decline in eGFR $\geq 40\%$ from baseline, or renal death in the FIGARO-DKD study



In a pre-specified pooled analysis of the FIDELIO-DKD and FIGARO-DKD studies, finerenone reduced the risk of the CV composite endpoint of time to CV death, non-fatal MI, non-fatal stroke or hospitalization for heart failure compared to placebo (HR 0.86 [95% CI 0.78; 0.95]). See Figure 5. The risk of the kidney composite endpoint of time to kidney failure, a sustained decrease in eGFR of 40% or more compared to baseline or renal death was also reduced with finerenone compared to placebo (HR 0.84 [95% CI 0.77; 0.92]), as was the composite endpoint of time to kidney failure, a sustained decrease in eGFR of 57% or more compared to baseline or renal death (HR 0.76 [95% CI 0.66; 0.88]). See Figure 5.

Figure 5: Cardiovascular and kidney composite outcomes in the pooled analysis of FIDELIO- DKD and FIGARO-DKD

Outcome	Finerenone (n=6498)		Placebo (n=6492)		Hazard ratio (95% CI)
	Number of patients with event (%)	Number of patients with event per 100 patient-years	Number of patients with event (%)	Number of patients with event per 100 patient-years	
Composite cardiovascular outcome	823 (12.7)	4.34	938 (14.4)	5.02	0.86 (0.78–0.95)
Death from cardiovascular causes	321 (4.9)	1.61	364 (5.6)	1.84	0.88 (0.75–1.02)
Non-fatal myocardial infarction	173 (2.7)	0.88	188 (2.9)	0.97	0.91 (0.74–1.12)
Non-fatal stroke	198 (3.0)	1.01	198 (3.0)	1.02	0.99 (0.82–1.21)
Hospitalisation for heart failure	255 (3.9)	1.31	325 (5.0)	1.69	0.78 (0.66–0.92)
eGFR 40% composite kidney outcome	848 (13.1)	4.80	995 (15.3)	5.66	0.84 (0.77–0.92)
Kidney failure	251 (3.9)	1.37	297 (4.6)	1.63	0.83 (0.70–0.98)
End-stage kidney disease	149 (2.3)	0.76	188 (2.9)	0.96	0.79 (0.64–0.98)
Sustained decrease in eGFR to <15 mL/min/1.73 m ²	192 (3.0)	1.05	237 (3.7)	1.30	0.80 (0.66–0.96)
Sustained $\geq 40\%$ decrease in eGFR from baseline	811 (12.5)	4.59	962 (14.8)	5.47	0.83 (0.76–0.92)
Renal death	2 (<0.1)	0.01	4 (<0.1)	0.02	0.54 (0.10–2.93)
eGFR 57% composite kidney outcome	356 (5.5)	1.95	465 (7.2)	2.56	0.76 (0.66–0.88)
Sustained $\geq 57\%$ decrease in eGFR from baseline	254 (3.9)	1.39	361 (5.6)	1.99	0.70 (0.59–0.82)

5.1.3.2 Heart Failure (LVEF ≥40%)

Firialta was investigated in a randomized, double-blind, placebo-controlled, multicenter phase III study, FINEARTS-HF. The effect of Firialta on cardiovascular outcomes was evaluated in adults with heart failure receiving either Firialta 10 mg, 20 mg, or 40 mg once daily, or placebo.

In FINEARTS-HF, patients were eligible with a diagnosis of heart failure with New York Heart Association (NYHA) class II–IV, ambulatory or hospitalized primarily for heart failure, and with documented left ventricular ejection fraction (LVEF) ≥40%. In addition, patients had an eGFR ≥25 mL/min/1.73m² and serum potassium ≤5.0 mmol/L at screening and randomization and were receiving background therapy, including diuretic treatment.

The primary endpoint was the composite of cardiovascular (CV) death and total (first and recurrent) heart failure events comprised of hospitalization for heart failure and urgent heart failure visits. The key secondary endpoints were total (first and recurrent) heart failure events, change from baseline to Month 6, 9, and 12 in Total Symptom Score (TSS) of the Kansas City Cardiomyopathy Questionnaire (KCCQ) (which quantifies heart failure symptom frequency and severity), improvement in NYHA class from baseline to Month 12, and first occurrence of the composite renal endpoint (sustained decrease in eGFR ≥50% relative to baseline over at least 4 weeks, or sustained eGFR decline to <15 mL/min/1.73m² or initiation of dialysis or renal transplantation), and time to all-cause mortality.

The study analyzed 6001 patients randomly assigned to receive either Firialta (N=3003) or placebo (N=2998), with a median follow-up duration of 2.7 years. The study included 3247 (54%) patients with a heart failure event in the past 3 months, including 1219 (20%) patients randomized during the hospitalization or within 7 days of discharge. At month 24, of the subjects treated with Firialta, 35% were treated with 40 mg once daily, 32% with 20 mg once daily, 12% with 10 mg once daily and 1% were on a treatment interruption. At any time during treatment, approximately 80% of the patients reached their target dose. After the end of study notification, vital status was obtained for 99.7% of patients. The trial population was 79% White, 17% Asian, and 1.5% Black. The mean age at enrollment was 72 years and 46% of patients were female. At baseline, the mean LVEF was 53%, with 64% of patients having an LVEF ≥50%, and 69% and 30% of patients were NYHA class II and III, respectively. Mean blood pressure was 129/75 mmHg while mean body mass index (BMI) was 30 kg/m². The median NT-proBNP was 1041 pg/mL, the mean eGFR was 62 mL/min/1.73m² with 48% of patients having an eGFR <60 mL/min/1.73m², and the median urinary albumin-to-creatinine ratio (UACR) was 18 mg/g. Atrial fibrillation was present for 38% of patients and 41% had diabetes mellitus. The majority of patients were on loop diuretics (87%), an angiotensin-converting enzyme inhibitor (ACEi) or angiotensin receptor blocker (ARB) (79%), or an angiotensin receptor neprilysin inhibitor (ARNI) (9%), and 14% were on sodium-glucose cotransporter 2 (SGLT2) inhibitors.

Firialta demonstrated superiority to placebo by significantly reducing the risk of the primary composite endpoint compared to placebo (RR 0.84, 95% CI 0.74-0.95, p=0.0072). See Table 6 and Figure 6 below. The effect was observed early with event curves separating from the first month and continuing to diverge throughout the study period. See Figure 6 below. Firialta also demonstrated superiority to placebo on the secondary endpoints of total heart failure events (RR 0.82, 95% CI 0.71-0.94, p=0.0062) and change from baseline to month 6, 9, and 12 in TSS of the KCCQ, indicative of improvements in symptom frequency and symptom burden (adjusted mean 1.56, 95% CI 0.79-2.34, p<0.0001). A numerical benefit of Firialta versus placebo was demonstrated for time to all-cause mortality, but this endpoint did not achieve statistical significance (HR 0.93, 95% CI 0.83-1.06). Prespecified secondary efficacy

endpoints are included in Table 6. The treatment effect for the primary and key secondary endpoints was consistent across all pre-specified subgroups, including sex, LVEF, NYHA class, eGFR, time since latest heart failure event, SGLT2 inhibitor therapy, and diabetes mellitus status.

Table 6: Analysis of the Primary and Secondary Endpoints (and their Individual Components for Time-to-Event Endpoints) in Phase III Study FINEARTS-HF

	Subjects with Heart Failure and LVEF $\geq 40\%$					
	Firialta 10 or 20 or 40 mg OD N=3003		Placebo N=2998		Treatment Effect Firialta vs Placebo	
Primary and Secondary Efficacy Endpoints	[Event Total] n (%) {Least Squares Mean}	Event Rate (100 pt-yr)	[Event Total] n (%) {Least Squares Mean}	Event Rate (100 pt-yr)	Hazard Ratio (HR) Odds Ratio (OR) Rate Ratio (RR) LS Mean difference (LSM) [95% CI]	p-value
Primary composite of CV death and total heart failure events	[1083] 624 (20.8%)	14.88	[1283] 719 (24.0%)	17.70	RR 0.84 [0.74, 0.95]	0.0072
Total heart failure events	[842] 479 (16.0%)	11.57	[1024] 573 (19.1%)	14.12	RR 0.82 [0.71, 0.94]	0.0062
CV death	242 (8.1%)	3.33	260 (8.7%)	3.59	HR 0.93 [0.78, 1.11]	-
Change from baseline in KCCQ-TSS	{7.99}	-	{6.43}	-	LSM 1.56 [0.79, 2.34]	<0.0001
Improvement in NYHA class	557 [†] (18.6%)	-	553 (18.4%)	-	OR 1.01 [0.88, 1.15]	0.9295*
Renal composite endpoint	75 (2.5%)	1.16	55 (1.8%)	0.85	HR 1.33 [0.94, 1.89]	-.**
Sustained eGFR decrease $\geq 50\%$	68 (2.3%)	1.05	51 (1.7%)	0.79	-	-

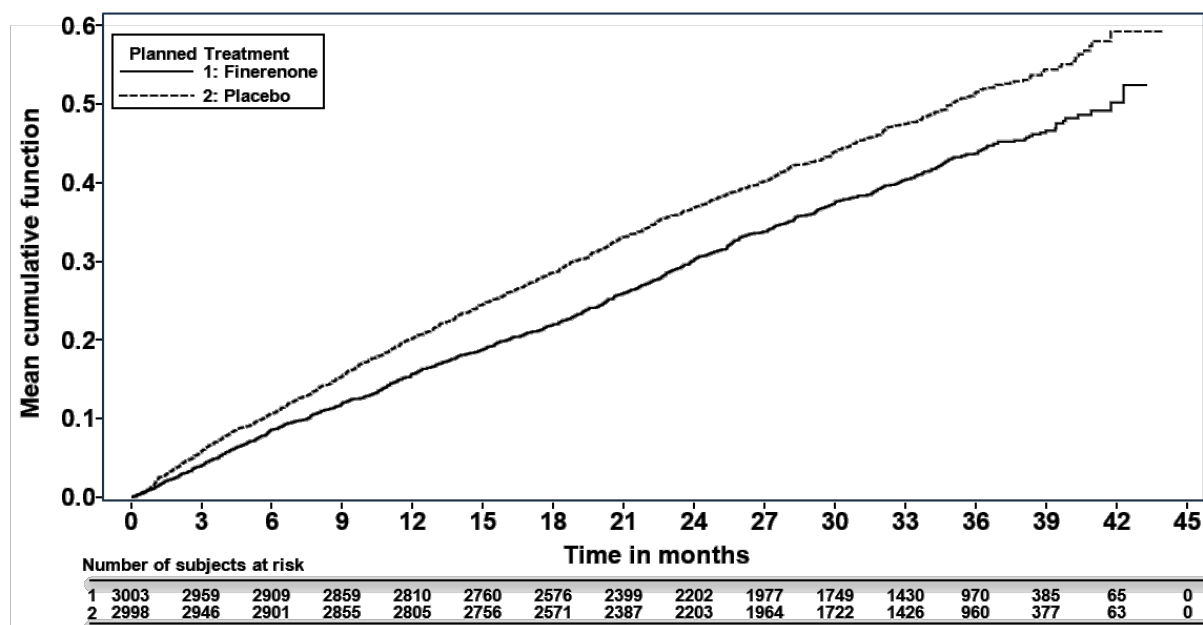
Sustained eGFR decline to <15 mL/min/1.73m ²	5 (0.2%)	0.08	2 (<0.1%)	0.03	-	-
Initiation of dialysis	2 (<0.1%)	0.03	2 (<0.1%)	0.03	-	-
Renal transplantation	0(0.0%)	0.00	0 (0.0%)	0.00	-	-
All-cause mortality	491 (16.4%)	6.71	522 (17.4%)	7.17	HR 0.93 [0.83, 1.06]	0.2794

* Not significant (testing procedure stopped)

** Endpoint not formally tested (preceding endpoint in testing procedure not significant)

† N=3002

Figure 6: Primary Composite Endpoint of CV Death and Total Heart Failure Events in Phase III Study FINEARTS-HF



5.2 Pharmacokinetic properties

5.2.1 Pharmacokinetic / Pharmacodynamic relationships

The concentration-effect relationship over time for UACR was characterized by a maximum effect model indicating saturation at high exposures. The model-predicted time to reach the full (99%) steady-state drug effect on UACR was 138 days. The pharmacokinetic (PK) half-life was 2-3 hours and PK steady state was achieved after 2 days, indicating timescale separation.

5.2.2 Absorption

Finerenone is almost completely absorbed after oral administration. Absorption is rapid with maximum plasma concentrations (C_{max}) appearing between 0.5 and 1.25 hours after tablet intake in the fasted state. The absolute bioavailability of finerenone is 43.5% due to first-pass metabolism in the gut-wall and liver. Finerenone is not a substrate of the efflux transporter P-gp *in vivo*. Intake with high fat, high calorie food increased finerenone AUC by up to 21%, reduced C_{max} by 19% to 23% and prolonged the time to reach C_{max} up to 2.5 hours. This is not clinically relevant. Therefore, finerenone can be taken with or without food (*see section 'Dosage and method of administration'*).

5.2.3 Distribution

The volume of distribution at steady state (V_{ss}) of finerenone is 52.6 L. The human plasma protein binding of finerenone *in vitro* is 91.7%, with serum albumin being the main binding protein.

5.2.4 Metabolism / Biotransformation

Approximately 90% of finerenone metabolism is mediated by CYP3A4 and 10% by CYP2C8. Four major metabolites were found in plasma, resulting from oxidation of the dihydropyridine moiety to a pyridine (M1a, M1b), subsequent hydroxylation of a methyl group (M2a) and formation of a carboxyl function (M3a). All metabolites are pharmacologically inactive.

5.2.5 Elimination / Excretion

The elimination of finerenone from plasma is rapid with an elimination half-life ($t_{1/2}$) of about 2 to 3 hours. Excretion of unchanged finerenone represents a minor route (<1% of dose in the urine due to glomerular filtration, < 0.2% in the feces). About 80% of the administered dose was excreted via urine and approximately 20% of the dose was excreted via feces, almost exclusively in the form of metabolites. With a systemic blood clearance of about 25 L/h, finerenone can be classified as a low clearance drug.

5.2.6 Linearity / Non-linearity

Finerenone pharmacokinetics are linear across the investigated dose range from 1.25 to 80 mg.

5.2.7 Additional information on special populations

5.2.7.1 Patients with renal impairment

Mild renal impairment (CLCR 60 - < 90 mL/min) did not affect finerenone AUC and C_{max}. Compared to subjects with normal renal function (CLCR ≥ 90 mL/min), the effect of moderate (CLCR 30 - < 60 mL/min) or severe (CLCR < 30 mL/min) renal impairment on AUC of finerenone was similar with increases by 34-36%. Moderate or severe renal impairment had no effect on C_{max} (*see section 'Dosage and method of administration'*).

Due to the high plasma protein binding, finerenone is not expected to be dialyzable.

5.2.7.2 Patients with hepatic impairment

There was no change in finerenone exposure in cirrhotic subjects with mild hepatic impairment (Child Pugh A) (*see section 'Dosage and method of administration'*).

In cirrhotic subjects with moderate hepatic impairment (Child Pugh B), finerenone mean AUC was increased by 38% and C_{max} was unchanged compared to healthy control subjects (*see section 'Dosage and method of administration'*).

There are no data in patients with severe hepatic impairment (Child Pugh C) (*see section 'Dosage and method of administration' and 'Special warnings and precautions for use'*).

5.2.7.3 Geriatric patients

Of the 2818 patients who received Firalta in the FIDELIO-DKD study, 58% of patients were 65 years and older, and 15% were 75 years and older. No overall differences in safety or efficacy were observed between these patients and younger patients.

Of the 3671 patients who received Firalta in the FIGARO-DKD study, 53% of patients were 65 years and older, and 14% were 75 years and older. No overall differences in safety or efficacy were observed between these patients and younger patients.

Of the 3003 patients who received Firalta in the FINEARTS study, 79% of patients were 65 years and older, and 43% were 75 years and older. No overall differences in safety or efficacy were observed between these patients and younger patients.

Elderly subjects (≥ 65 years of age) exhibited higher finerenone plasma concentrations than younger subjects (≤ 45 years of age), with mean AUC and C_{max} values being 34% and 51% higher in the elderly (*see section 'Dosage and method of administration'*).¹³

Population-pharmacokinetic analyses did not identify age as a covariate for finerenone AUC or C_{max}.

5.2.7.4 Sex

Sex had no effect on the pharmacokinetics of finerenone (*see section 'Dosage and method of administration'*).

5.2.7.5 Body Weight

Population-pharmacokinetic analyses identified body weight as a covariate for finerenone C_{max} and AUC. C_{max} and AUC of subjects with a body weight below 57 kg were estimated to be, on average, 52% and 30% higher, respectively, and of subjects with a body weight above 122 kg, 32% and 20% lower, respectively, compared to a subject between 57 and 122 kg. Dose adaptation based on body weight is not warranted (*see section 'Dosage and method of administration'*).

5.2.7.6 Race/Ethnicity

Population-pharmacokinetic analyses in patients demonstrated no clinically relevant difference in finerenone exposure between Asian and White patients (*see section 'Dosage and method of administration'*).

5.2.7.7 Smoking status

Finerenone is not metabolized by an enzyme that is inducible by tobacco smoke (*see section 'Dosage and method of administration'*).

5.3 Preclinical safety data

5.3.1 Embryotoxicity / Teratogenicity

In the embryo-fetal toxicity in rats, finerenone resulted in reduced placental weights and signs of fetal toxicity including reduced fetal weights and retarded ossification at the maternal toxic dose of 10 mg/kg/day corresponding to an AUC_{unbound} of at least 7 times that in humans. At 30 mg/kg/day, the incidence of visceral and skeletal variations was increased (slight edema, shortened umbilical cord, slightly enlarged fontanelle) and one fetus showed complex malformations including a rare malformation (double aortic arch) at an AUC_{unbound} of about 10 times that in humans at the 40 mg dose and about 25 times that in humans at the 20 mg dose. The doses free of any findings (low dose in rats, high dose in rabbits) provided safety margins of 4 to 13 times for AUC_{unbound}. Therefore, the findings in rats do not indicate an increased concern for fetal harm (*see section 'Fertility, pregnancy and lactation'*).

When rats were exposed during pregnancy and lactation in the pre- and postnatal developmental toxicity study, increased pup mortality and other adverse effects (lower pup weight, delayed pinna unfolding) were observed at about 2 or 4 times the AUC_{unbound} expected in humans at the dose of 40 mg and 20 mg, respectively. In addition, the offspring showed slightly increased locomotor activity, but no other neurobehavioral changes starting at about 2 or 4 times the AUC_{unbound} expected in humans at the dose of 40 mg and 20 mg, respectively. The dose free of findings provided a safety margin of about 2 for AUC_{unbound} for the 20 mg dose and is in the therapeutic range for the 40 mg dose. The increased locomotor activity in offspring may indicate a potential risk for the fetus. In addition, because of the findings in pups, a risk for the nursing infant cannot be excluded (*see section 'Special warnings and precautions for use' and 'Fertility, pregnancy and lactation'*).

5.3.2 Reproduction toxicity

Male fertility was not affected by Firialta (*see section 'Fertility, pregnancy and lactation'*).

Finerenone caused reduced female fertility (decreased number of corpora lutea and implantation sites) as well as signs of early embryonic toxicity (increased post-implantational loss and decreased number of viable fetuses) at about 9 times the human AUC_{unbound} for the 40 mg dose and at about 21 times the human AUC_{unbound} for the 20 mg dose. In addition, reduced ovarian weights were found at about 7 times the humans AUC_{unbound} for the 40 mg dose and at about 17 times the human AUC_{unbound} for the 20 mg dose. No effects on female fertility and early embryonic development were found at 4 times the human AUC_{unbound} for the 40 mg dose and at 10 times the human AUC_{unbound} for the 20 mg dose. Therefore, the findings in female rats are of little clinical relevance (*see section` Fertility, pregnancy and lactation`*).

5.3.3 Genotoxicity and carcinogenicity

Finerenone was non-genotoxic in an in vitro bacterial reverse mutation (Ames assay), the in vitro chromosomal aberration assay and the in vivo micronucleus assay.

In 2-year carcinogenicity studies, finerenone did not show a carcinogenic potential in Wistar rats as well as CD1 mice. In male mice, finerenone resulted in an increase in Leydig cell adenoma at doses representing 10 to 26 times the AUC_{unbound} in humans. A dose representing 7 times the AUC_{unbound} for the 40 mg dose and 17 times the AUC_{unbound} in humans for the 20 mg dose did not cause any tumors. Based on the known sensitivity of rodents to develop these tumors and the pharmacology-based mechanism at supratherapeutic doses as well as adequate safety margins, the increase in Leydig cell tumors in male mice is not clinically relevant.

6. PHARMACEUTICAL PARTICULARS

6.1 List of excipients

Tablet core:

Cellulose microcrystalline (E 460)

Croscarmellose sodium

Hypromellose 5 cP (E 464)

Lactose monohydrate

Magnesium stearate (E 470b)

Sodium laurilsulfate (E 487)

Tablet coating:

Ferric oxide red (E 172) (Firialta 10 mg film-coated tablet)

Ferric oxide yellow (E 172) (Firialta 20 mg film-coated tablet)

Hypromellose 5 cP (E 464)

Talc (E 553 b)

Titanium dioxide (E 171)

6.2 Incompatibilities

N/A

6.3 Shelf life

Please refer to labels

6.4 Special precautions for storage

Store below 30 °C

6.5 Nature and contents of container

PVC/PVDC-Aluminium transparent calendarised blisters with 14 film-coated tablets. Pack sizes of 14, 28 or 98 film-coated tablets

Not all pack sizes may be marketed.

6.6 Instructions for use / handling

Any unused medicinal product or waste material should be disposed of in accordance with local requirements.

MANUFACTURER

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